Cohabitation timing and childbearing or current pregnancy among married adolescents in Bangladesh

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Introduction

Adolescent childbearing is an important reproductive-health concern in Bangladesh. Timing of first cohabitation also determines how long childbearing could have occurred before interview. [1,2]

Objective: Describe live-birth experience or current pregnancy among ever-married adolescents, and examine conditional associations with cohabitation timing and partner characteristics.

Methods

Data: 2022 Bangladesh Demographic and Health Survey (BDHS), ages 15–19. [1]

Outcome: At least one live birth or currently pregnant at interview. This is a proportion, not a birth rate.

Regression: Survey-weighted logistic model with current age, respondent education, official wealth quintile, cohabitation age, husband education, signed spousal age gap, residence and division. Work was examined separately.

Design: V005/1,000,000 weights, V021 PSUs and V023 strata. Domain estimation retained the full design. Taylor variance and t-based 95% intervals. [3]

Sample selection

2,449 ever-married adolescents (descriptive sample)
1,635 received the long questionnaire
1,603 were currently married
1,601 had complete regression information

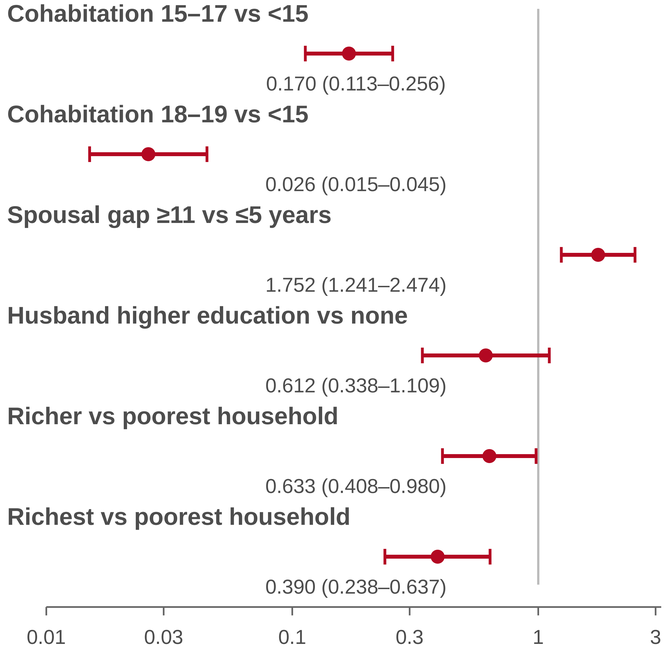
Exclusions: 814 short questionnaires, 32 formerly married respondents and 2 unknown husband-education values. Questionnaire assignment is distinct from item nonresponse.

Results

59.7%

Had ≥1 live birth or were currently pregnant
95% CI 57.6–61.7%
Ever-married adolescents aged 15–19 (n = 2,449)

Selected adjusted odds ratios (n = 1,601)



Model checks and weighting audit

R/Python estimates agreed. Maximum selected-coefficient change after deleting one PSU: 0.286 SE.

Held-out-PSU AUC: 0.793. Calibration slope: 0.885. Possible misspecification (exploratory p ≈ 0.0205).

Constant long-form selection factor (45/30): estimates and Taylor SEs unchanged.

Discussion

Later first cohabitation was associated with lower adjusted odds of childbearing or current pregnancy. A spousal age gap of at least 11 years was associated with higher odds. Higher husband education remained inconclusive after expanded adjustment.

Interpretation depends on the observation window: current age and time since cohabitation strongly shape these cumulative outcomes. Cohabitation at ages 18–19 occurs only among respondents already aged 18–19.

Limitations: Cross-sectional data, selection into marriage, interview-time covariates and limited age/cohabitation overlap constrain causal interpretation. Results do not estimate the effect of delaying marriage or a marriage law.

Conclusion

Cohabitation timing and spousal age gap show reproducible conditional associations in this population.

Interpret findings as associations among married adolescents. Further work should address timing and model specification before causal or predictive use.

Acknowledgements

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References

[1] NIPORT & ICF. Bangladesh Demographic and Health Survey 2022. 2024. Sections 1.2–1.3, Table 3.1.2, Appendix A.4.
[2] Bongaarts J. The proximate determinants of fertility. Technology in Society. 1987;9:243–260.
[3] Lumley T. Analysis of complex survey samples. Journal of Statistical Software. 2004;9(8):1–19.